

Spring Boot RestTemplate Consumer Producer Communication Workbook

Task 1: Student Service and Course Service Communication

Objective:

Develop two Spring Boot applications that communicate using RestTemplate.

APPLICATION 1: COURSE SERVICE (PRODUCER)

Step 1: Create Spring Boot Project

Dependencies: Spring Web, Spring Data JPA, MySQL Driver, Swagger/OpenAPI, Lombok

Step 2: Create Database

Database: course_db

Table: courses

Step 3: Create Entity

Fields: courseId, courseName, trainerName, duration, fee

Step 4: Create Repository Layer

CourseRepository extends JpaRepository

Step 5: Create Service Layer

Methods:

addCourse()

getAllCourses()

getCourseById()

updateCourse()

deleteCourse()

Step 6: Create Controller Layer

POST /courses

GET /courses

GET /courses/{id}

PUT /courses/{id}

DELETE /courses/{id}

Step 7: Use ResponseEntity

Step 8: Create Custom Exception

CourseNotFoundException

Step 9: Create Global Exception Handler

@RestControllerAdvice

Step 10: Swagger Integration

@Tag

@Operation

@ApiResponse

APPLICATION 2: STUDENT SERVICE (CONSUMER)

Step 1: Create Spring Boot Project

Step 2: Create Database

Database: student_db

Table: students

Step 3: Create Entity

Fields:

studentId

studentName

email

courseId

Step 4: Create Repository Layer

StudentRepository extends JpaRepository

Step 5: Create DTO Classes

CourseDTO

StudentCourseResponseDTO

Step 6: Configure RestTemplate Bean

Step 7: Create Service Layer

addStudent()

getAllStudents()

getStudentById()

updateStudent()

deleteStudent()

getStudentWithCourse()

Step 8: RestTemplate Flow

Fetch Student

Call Course Service

Receive CourseDTO

Combine Data

Return Response DTO

Step 9: Create Controller APIs

POST /students

GET /students

GET /students/{id}

PUT /students/{id}

DELETE /students/{id}

GET /students/course/{id}

Step 10: Exception Handling

StudentNotFoundException

HttpClientErrorException

Concepts Covered:

Controller Layer

Service Layer

Repository Layer

DTO Layer

Entity Layer

RestTemplate

ResponseEntity
Swagger
Global Exception Handling
MySQL
JPA

Practice Assignments:
Employee ↔ Department
Order ↔ Product
Appointment ↔ Doctor
Account ↔ Customer
Movie ↔ Theatre
Hotel ↔ Room
Flight ↔ Passenger
E-Commerce Order ↔ Product + Inventory